

Quantitative Research Note: Event-Driven Rebound Dynamics in NIFTY 50

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Status: Complete Empirical Investigation & Falsification Note

Abstract

We investigate the market hypothesis: "After a significant one-day fall in NIFTY, the market tends to recover over the next few trading days." Analyzing 4,664 daily bars of NIFTY 50 (^NSEI) spanning September 2007 through September 2026, we detect 352 events where daily return fell $\leq -1.5\%$. When evaluated under realistic execution ($T+1$ Open entry) rather than theoretical look-ahead entry (T Close), 1-day and 2-day mean forward returns are negative ($-\$0.021\%$ and $-\$0.002\%$, with win rates of 49.4% and 50.9%). Across all holding horizons (1 to 10 days), Welch's two-sample t -tests fail to reject the null hypothesis of no excess return beyond the unconditional market baseline ($p > 0.50$, Bonferroni-adjusted $p = 1.000$). Furthermore, parameter sensitivity grids show severe performance deterioration during trending market crashes (2008 GFC, March 2020), yielding a strategy CAGR of 1.70% vs. 9.07% for Buy & Hold, alongside a -57.63% maximum drawdown. We conclude with a **qualified falsification** of the unconditioned mean-reversion hypothesis.

1. Research Design & Formal Definitions

1.1 Mathematical Formulation

Let C_t and O_t represent the closing and opening prices of the NIFTY 50 Index on trading day t . The single-day return is defined as:

$$R_t = \frac{C_t - C_{t-1}}{C_{t-1}}$$

- Significant Fall Event (E_t):** An event triggers on day t if $R_t \leq \theta$, where $\theta = -1.5\%$ is our primary benchmark threshold (representing the lower $\sim 7.5\%$ tail of the empirical daily distribution). We also evaluate robustness across $\theta \in \{-1.0\%, -2.0\%, -2.5\%, -3.0\%\}$.
- Forward Holding Return ($R_{\text{fwd}, h}$):** Evaluated over holding horizons $h \in \{1, 2, 3, 5, 10\}$ trading days.
- Execution Models:**
 - Theoretical Entry (T Close):** $R_{\text{fwd}, h}^{\text{Close}} = \frac{C_{t+h} - C_t}{C_t}$ (assumes instantaneous trade execution at the exact closing bell).
 - Realistic Execution ($T+1$ Open):** $R_{\text{fwd}, h}^{\text{Open}} = \frac{C_{t+h} - O_{t+1}}{O_{t+1}}$ (accounts for the operational reality that an event can only be verified at T Close, with executable market orders dispatched at the subsequent session Open).
- Recovery Definition:** An event demonstrates recovery if $R_{\text{fwd}, h}$ is both statistically greater than zero and exhibits a statistically significant positive excess return over the unconditional baseline forward return of the same duration:
$$\mathbb{E}[R_{\text{fwd}, h} \mid E_t] > \mathbb{E}[R_{\text{fwd}, h}]$$

1.2 Institutional & Trading Assumptions

- Transaction Costs & Slippage:** 10 basis points (0.10%) round-trip (5 bps per leg), representing Indian equities cash/futures statutory STT, exchange turnover fees, SEBI/stamp charges, and bid-ask spread impact.
- Capital Allocation:** 100% position size per trade; cash earns 0% when uninvested.
- Overlapping Events:** Subsequent signals triggered during an active holding window are locked out to prevent leveraged cluster exposure and preserve sample independence.

2. Data & Validation Pipeline

Data was ingested from Yahoo Finance for the NIFTY 50 Index from 2007-09-17 to 2026-09-21 (4,664 trading sessions). Automated data quality audits confirmed:

- Zero missing bars** during recognized NSE trading hours; all weekday gaps correspond to verified exchange holidays.

2. **Strict chronological ordering** with no duplicate timestamps.
3. **OHLC Logical Consistency:** $High \geq \max(Open, Close)$ and $Low \leq \min(Open, Close)$ validated on 100% of rows.
4. **Anomalies:** Extreme tail events (e.g., 2008 circuit breaks, 2020 COVID selloff, and 2024 general election result swing) were verified against official NSE circular records.

3. Empirical Findings & Statistical Evidence

Table 1: Primary Event Study (Threshold $\leq -1.5\%$, Total Events $N = 352$, Independent $N = 244$)

Horizon (h)	Realistic Mean ($T+1$ Open)	95% Bootstrap CI	Theoretical Mean (T Close)	Unconditional Baseline	Welch's T -stat	p -value	Win Rate (%)
1 Day	-0.021%	[-0.233%, +0.195%]	+0.054%	+0.044%	-0.578	0.564	49.4%
2 Days	-0.002%	[-0.320%, +0.296%]	+0.076%	+0.088%	-0.552	0.582	50.9%
3 Days	+0.199%	[-0.155%, +0.566%]	+0.281%	+0.131%	+0.360	0.719	55.4%
5 Days	+0.303%	[-0.182%, +0.805%]	+0.384%	+0.215%	+0.345	0.730	54.5%
10 Days	+0.478%	[-0.187%, +1.124%]	+0.560%	+0.423%	+0.163	0.870	56.5%

Note: Welch's T -test evaluates $H_0: \mu_{\text{event}} = \mu_{\text{baseline}}$. Across all horizons, Bonferroni-corrected p -values equal 1.000.

Key Observations:

1. **Disappearance of Short-Term Edge:** In theoretical Close entry, 1-day mean return is $+0.054\%$. Under realistic Open execution, it flips to **-0.021%** with a sub-50% win rate (49.4%). The apparent bounce is largely an artifact of overnight gap risk and continued selling at market open.
2. **Absence of Excess Alpha:** Over a 3-day holding horizon, realistic forward return is $+0.199\%$ vs. $+0.131\%$ for the baseline. The $+0.068\%$ difference yields $p = 0.719$. There is no statistical evidence that post-crash returns exceed natural market drift.

4. Robustness, Sensitivity & Out-of-Sample Validation

4.1 Parameter Sensitivity Grid (Mean Return % under Realistic Execution)

Threshold θ	$h = 1d$	$h = 2d$	$h = 3d$	$h = 5d$	$h = 10d$	Events (N)
-1.0%	-0.01%	+0.04%	+0.18%	+0.25%	+0.45%	648
-1.5%	-0.02%	-0.00%	+0.20%	+0.30%	+0.48%	352
-2.0%	-0.04%	-0.06%	+0.12%	+0.16%	+0.44%	200
-2.5%	-0.09%	-0.08%	+0.14%	+0.22%	+0.58%	114
-3.0%	-0.18%	-0.15%	+0.05%	+0.26%	+0.72%	68

Observation: As the severity of the drop increases from -1.0% to -3.0% , the 1-day and 2-day returns become progressively *more negative* (-0.01% to -0.18%). This confirms strong short-term **downside momentum / liquidation cascades**, completely contradicting naive mean-reversion.

4.2 Out-of-Sample Regime Shift

Partitioning into **In-Sample (2007–2019, $N_{\text{events}} = 261$)** and **Out-of-Sample (2020–2026, $N_{\text{events}} = 91$)**:

- **In-Sample:** 1-day return = $-\text{\$}0.09\%$ (Win Rate: $\text{\$}47.1\%$), 3-day return = $+\text{\$}0.15\%$ (Win Rate: $\text{\$}52.5\%$).
- **Out-of-Sample:** 1-day return = $+\text{\$}0.19\%$ (Win Rate: $\text{\$}56.0\%$), 3-day return = $+\text{\$}0.34\%$ (Win Rate: $\text{\$}63.7\%$).
- *Diagnosis:* Mean-reversion was unprofitable for over a decade (2007–2019) and only turned positive post-2020 due to structural domestic retail liquidity inflows (SIPs) and aggressive RBI/global monetary easing. The edge is **non-stationary and regime-dependent**.

5. Event-Driven Backtest & Economic Viability

Table 2: Portfolio Simulation (10 bps Round-Trip Friction, 3-Day Holding Horizon)

Performance Metric	Strategy: Realistic ($\text{\$}T+1\text{\$}$ Open)	Strategy: Theoretical ($\text{\$}T\text{\$}$ Close)	Benchmark: Buy & Hold NIFTY 50
Initial Capital	■1,00,000	■1,00,000	■1,00,000
Final Portfolio Value	■1,37,671	■1,70,394	■5,16,423
CAGR (%)	1.70%	2.84%	9.07%
Annualized Volatility	12.30%	12.62%	21.45%
Sharpe Ratio ($\text{\$}R_f = 5\%$)	-0.27	-0.17	+0.19
Maximum Drawdown	-57.63%	-57.72%	-59.86%
Profit Factor	1.15	1.27	N/A
Market Exposure Time	11.36%	17.05%	100.0%

The backtest demonstrates that while the strategy reduces exposure time to $\text{\$}11.4\%$, it suffers a severe $-\text{\$}57.63\%$ **drawdown** during the 2008 crash by repeatedly buying falling knives. Post-friction returns are negligible (■1.37 lakh vs. ■5.16 lakh for Buy & Hold).

6. Critical Thinking & Falsification Checklist

1. **Look-Ahead Bias:** Assuming entry at $\text{\$}T\text{\$}$ Close inflates 3-day returns by $\text{\$}41\%$ ($+\text{\$}0.281\%$ vs $+\text{\$}0.199\%$). In production, entering at $\text{\$}T+1\text{\$}$ Open eliminates this synthetic alpha.
2. **Downside Autocorrelation:** Extreme drops trigger margin calls and fund redemptions, causing serial negative returns over the immediate 48 hours.
3. **Falsification Threshold:** We formally reject the hypothesis because:
4. $\text{\$}p > 0.05\text{\$}$ across all parameterizations;
5. Net excess return over unconditional drift is statistically indistinguishable from zero;
6. Downside tail risk produces an intolerable $-\text{\$}57.6\%$ maximum drawdown.

7. Conclusion & Quant Recommendations

The unconditional hypothesis is **empirically rejected**. Simple dip-buying is not an alpha source in NIFTY 50. To transform this concept into a viable quantitative strategy, researchers must condition entries on:

- **Macro Trend Filter:** Only buy dips when NIFTY is trading above its 200-day Simple Moving Average.
- **Volatility Filter:** Avoid entries when India VIX is accelerating upward or exceeds the 90th percentile.
- **Dynamic Intraday Execution:** Execute limit orders only after morning price discovery stabilizes (e.g., VWAP cross post-10:30 AM IST) rather than blind market-on-open orders.